# Three ways to learn the DuplexCascade micro-turn protocol: supervised fine-tuning, self-distillation, and a policy companion

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***Abstract*—Full-duplex spoken dialogue requires a conversational model to do two things at once: produce answer content and control turn-taking—when to speak, when to stop, when to acknowledge, when to yield. The DuplexCascade paradigm encodes both in one language model by interleaving answer text with control tokens over *micro-turns*. Learning that protocol is cheap, but the fine-tune is destructive: in our Brazilian-Portuguese implementation the plain-SFT model that learns the tags with 98% accuracy simultaneously loses most of its function calling (100%→55.6%) and general instruction following (100%→41.7%). This paper studies three ways to obtain the protocol without that cost, on one shared pt-BR data pipeline, and evaluates them with one shared harness. (i) SFT is the original recipe, kept as the baseline and as the fastest deployable model. (ii) Self-distillation splits the objective by token position: weighted cross-entropy on the tags exactly as before, plus a per-token forward-KL to the frozen base model on every content position. It needs neither reinforcement learning nor on-policy rollouts, and with an offline teacher cache it costs no more than SFT; it matches the protocol (96.4% vs. 98% tag accuracy) while holding function calling, general instruction-following and knowledge benchmarks at base level. (iii) An MTP-style protocol companion separates the skills instead of balancing them: a 0.8B sidecar predicts *only* the control stream while an untouched big model writes the answer, reaching 95.6% tag-stream accuracy with 0.97/0.99 F1 on the two critical controller decisions (barge-in interruption and backchannel) and supporting instruction-conditioned interrupts. All three are then evaluated in *free generation* with a Full-Duplex-Bench harness, and against the upstream English DuplexCascade release re-measured on the same harness, which reproduces its own published turn-taking numbers (ATA 0.896 vs. 0.858 published). All three pt-BR variants beat that published accuracy (ATA 0.955–0.973). The comparison also exposes what teacher-forced accuracy cannot see: the distill model, despite the best capability retention, has the weakest interruption relevance (1.59 vs. 3.56 and 4.36) and commits to the floor $13\times$ more slowly (3.29 s vs. 0.25 s), because the KL anchors the teacher's *generation style* along with its capabilities. We further show that the evaluated input stream must match the training stream—on the same model, one ablation arm more than triples pause-handling TOR and another destroys interruption recovery, and the two axes are separable—and that one FDB dimension is unanswerable on this data by construction, since the system-backchannel tag occurs zero times in training.**

***Index Terms*—full-duplex dialogue, turn-taking, self-distillation, catastrophic forgetting, parameter-efficient fine-tuning, MTP, VAD-free, evaluation**

## I. Introduction

SPOKEN dialogue systems are moving from turn-based ("push-to-talk") interaction to *full-duplex* conversation, where both parties may speak at any time. The DuplexCascade paper [1] established that a language model can learn the full-duplex protocol directly: its training data interleaves user and assistant *micro-turns* and supervises a set of control tokens ("user is speaking", "user finish speaking", "user interruption", "user backchannel", "user is thinking", "system backchannel") so that the same model that writes the answer also emits the control stream. The model must therefore be fine-tuned—deeply— for the turn-taking skill.

Learning the protocol is easy: the supervision is cheap, synthetic and deterministic. The fine-tune is not. In our pt-BR implementation, the plain-SFT model that learns the tags with 98% accuracy simultaneously loses most of its ability to call functions (100%→55.6%) and to follow general instructions (100%→41.7%). This is catastrophic forgetting in its most practical form: the model becomes a good turn-taker and a bad assistant.

This paper studies three answers to that problem, all built on the same pt-BR duplex pipeline and all evaluated with the same harness:

- **SFT** (Section IV)—the original recipe: QLoRA on the duplex micro-turn data with weighted, position-masked cross-entropy. It remains the best raw turn-taker and the fastest to deploy, and it is the baseline the other two are measured against.
- **Self-distillation** (Section V)—the observation that the new skill and the capabilities to preserve are *separable by token position*. The tags live at supervised positions; the content lives at the rest. Splitting the objective accordingly—weighted CE on the tags, per-token forward-KL to the frozen base everywhere else— retains the capabilities without RL and without rollouts.
- **An MTP-style protocol companion** (Section VI)—the inverse decomposition: a 0.8B sidecar learns *only* the protocol and drives an untouched big model, which is the same division of labour as a Multi-Token-Prediction

head except that the predicted vocabulary is the control protocol rather than the next answer token.

*Is the protocol actually used?* The protocol metric these methods are usually compared on is teacher-forced: at each supervised position the model is handed the correct prefix and only its next-token choice is scored. That measures whether the protocol was *learned*; it says nothing about whether the model *decides* correctly when generating on its own, which is what a deployed duplex system needs. We therefore evaluate all three, plus the untrained base as a control, with a free-generation harness modelled on Full-Duplex-Bench (FDB) [3], and—to anchor the comparison—we re-measure the upstream English DUPLEXCASCADE release with the same harness instead of quoting it. That reference lands on its own published numbers, which is the strongest available check that the harness measures what the benchmark measures.

Four findings follow from pairing the two evaluations, and they are the contribution of Sections VII and VIII:

1) **All three approaches reach the protocol; they differ in what it costs.** SFT and the companion reach the best turn-taking accuracy (ATA 0.971/0.973), distill is close (0.955) and all three exceed the published 0.858—but the companion also has the best interruption relevance (4.36) and the fastest commitment among the decomposed designs, while SFT is an order of magnitude faster to stop and respond (0.035 s vs. 0.066 s and 0.351 s).
2) **Teacher forcing hides a generation-style cost.** The distill model has the best capability retention and the worst free-generation reply quality (interruption relevance 1.59) with a $13\times$ slower floor commitment. The content-KL anchors the teacher's generation style together with its capabilities, and the teacher occasionally autocompleted the user's turn—an artifact teacher-forced accuracy (96.4%) cannot detect.
3) **The input stream is an experimental variable.** The models are trained on 1–7-token ASR pieces and $\sim$10-word turns. Two single-factor ablations show the axes are separable: multi-sentence turns more than triple pause-handling TOR (0.080→0.307) while whole-sentence pieces destroy interruption recovery (0.995→0.000). Any free-generation protocol benchmark must therefore report the input distribution it used.
4) **A metric can be unanswerable by construction.** No model here can score the FDB backchannel dimension, because the tag it requires (`<|system backchannel|>`) occurs zero times in the training data. The upstream release, trained elsewhere, scores it—so the column measures a data gap, not a capability gap.

## II. BACKGROUND AND RELATED WORK

### A. DuplexCascade and the micro-turn protocol

DuplexCascade [1] keeps the intelligence of a text LLM by not modelling audio: a streaming ASR feeds text to the LLM, which emits interleaved answer text and control tokens, and a streaming TTS speaks the text. Turn-taking is *VAD-free* in the sense that no separate voice-activity detector decides who speaks: the LLM itself emits control tokens based on the incoming text stream. The dialogue is decomposed into *micro-turns*—short interleaved user and assistant fragments—so that the system can react before an utterance is complete. We adopt this setting exactly, and our contributions are about *how the protocol is learned* and *how it is measured*, not about the architecture.

### B. Catastrophic forgetting and off-policy fine-tuning

Sequential SFT is inherently off-policy: the model learns to imitate expert actions on a fixed offline distribution, which prior work shows causes severe forgetting when adapting to a new task or domain [5]–[7]. The standard remedy is to make the signal on-policy with reinforcement learning and an explicit reward [8], [9]. Rewards are often unavailable for a protocol skill, and on-policy RL (PPO/GRPO) is expensive and brittle. This motivates the distillation-based approaches we compare against plain SFT.

### C. Self-distillation

*Self-Distillation Fine-Tuning* (SDFT) [2] obtains on-policy benefits from demonstrations alone: the model acts as a *teacher* conditioned on a demonstration and as a *student* conditioned only on the query, and training minimizes a KL between the two on the student's own trajectories. It reduces forgetting substantially relative to SFT, but still needs on-policy rollouts (roughly $4\times$ SFT wall-clock). Related on-policy-distillation work reaches similar conclusions from different directions [9]–[13]. Our self-distillation variant (Section V) is the degenerate case of this family in which the demonstration is the training sequence itself, so the teacher is simply the frozen base and no rollouts are required.

### D. Parameter-efficient fine-tuning

We use LoRA [14] and its quantized variant [15] throughout, both for the 4B duplex models and for the 0.8B companion, with the toolchain of [18]. PEFT matters here for a second reason: because the student is *base* + *adapter*, the teacher forward pass is obtained by temporarily disabling the adapters, so the distillation teacher costs no extra VRAM and no second checkpoint.

### E. Multi-token prediction and VAD-free turn-taking

Multi-Token Prediction heads [16] predict several future tokens from a shared trunk with a lightweight module. Our companion (Section VI) borrows the *architectural* idea—a small module running alongside a large frozen model—but predicts a different vocabulary: the duplex protocol. It is therefore closer in spirit to a speculative or auxiliary head than to a pretraining objective. The companion is trained on the same data and with the same distill objective as Section V, which makes the two directly comparable.

## III. Shared infrastructure: the pt-BR micro-turn pipeline

All three approaches consume one dataset and target one protocol vocabulary, so we describe them once.

### A. Protocol vocabulary and supervision

A dialogue is a strict alternation of user and assistant *items*, each rendered in ChatML and closed with `<|im_end|>`; assistant items may carry one control tag at their start. The tags are `<|user is speaking|>`, `<|user finish speaking|>`, `<|user is thinking|>`, `<|user interruption|>`, `<|user backchannel|>`, `<|system backchannel|>`, plus the input-only marker `<|no voice|>` used for silence. Following the paper, loss is computed only on assistant micro-turns and each tag carries a task-specific weight: `user finish speaking` $\times 10$, `user interruption` $\times 5$, `system backchannel` $\times 3$, `user backchannel` $\times 2$, `user is speaking` and `user is thinking` $\times 1$.

### B. Data: source dialogues and micro-turn construction

Two generations of source dialogues exist, and the difference matters for evaluation (Section VII-C). The first generator instructed *"respostas curtas do Assistente. Nada de textos longos"*, and the chunker fixed every assistant micro-turn at exactly 10 tokens; the resulting models spoke only in $\sim$10-token bursts. The continuation generator (`build_long_dialogues.py`) mixes *short* (weight 6), *long* (2) and *mixed* (2) styles, so a fraction of assistant turns are 2–4-sentence replies up to 220 words, and system chunking samples 10–48 tokens instead of a constant. Both models in this paper train on the continuation data; the companion trains on the same dialogues.

The deterministic micro-turn builder then converts each dialogue into items with six phenomena: randomized micro-turn length (user chunks 1–7 tokens), natural pauses (1–5 `<|no voice|>` items), user interruptions ($p = 0.30$, with the first interrupting chunk labelled `user interruption` and later ones `user is speaking`), user backchannels ($p = 0.01$, from a pt-BR interjection list), system backchannels (opt-in, $p = 0.20$ at sentence-final user chunks) and user thinking pairs (1–20 `<|user is thinking|>` items closing each completed system turn). Strict user/assistant alternation is asserted. The generator *oversamples interruptions* in FDB's sense in one respect worth noting: it never samples the position where an interruption begins, so we treat its 0.30 rate as an augmentation strength rather than a label prior.

### C. Measured turn sizes

Because FDB's turn-taking gates are *durations*, both sides' length distributions are load-bearing. Measured over the training data (3,000 sequences):

- **User pieces**: mean **2.9 words** per piece (median 3, p90 5), arriving as 3.7 pieces per turn—so a user turn is $\sim$10.5 words and contains $\sim$1.5 sentence-final pieces.
- **Assistant content bursts**: mean 6.1 words but *median 5, p90 12, max 37*, with 50% of bursts at $\leq$5 words and 1% above 20. The heavy right tail is the intended effect of the variable chunking and the longer source dialogues.

The consequence for evaluation is a mixture rather than a mean: a model trained on longer bursts crosses FDB's "formal turn" threshold ($\geq$1 s) with fewer words, so the assistant-side length distribution moves the metric being reported. We return to this in Section VII-C and report both sides' lengths with the results.

### D. Runtime

The live stack is four components: `llama.cpp` serving the fine-tuned GGUF, `faster-whisper` for streaming ASR, `pocket-tts` for streaming speech synthesis, and a bridge that wires them and exposes a browser UI. Two serving details are mandatory and were sources of silent failure: the server must be started with `-sp` so the special tokens are returned by the completions endpoint at all, and the bridge must post to the raw completions endpoint rather than the chat endpoint, because the chat template mangles `<|...|>` tokens appearing in assistant history.

## IV. Approach A: supervised fine-tuning (the original recipe)

### A. Method

QLoRA with $r$=16, $\alpha$=16, dropout 0, applied to all linear modules, plus `modules_to_save=["embed_tokens","lm_head"]` so that the seven new special-token embeddings are trained. The base is `Qwen/Qwen3-4B-Instruct-2507`. The objective is a weighted, position-masked cross-entropy: labels are shifted by one, every user position and every ChatML header is masked to $-100$, and each supervised tag token carries the weight of Section III. Training runs in two stages: an initial fine-tune, then a continuation fine-tune started from the *merged* model on a short+long response mixture. Starting the second stage from the merged model rather than the base is required—from the base, the learned special-token embeddings and the turn-taking pattern are re-randomized and lost.

### B. What it costs

SFT is the fastest model to serve and, as Section VIII shows, the best raw turn-taker of the three. Its cost is the capability regression of Section VIII: it is the baseline that motivates the other two approaches, not a straw man.

## V. Approach B: self-distillation by position

### A. The observation

The "new skill" (the turn-taking protocol) and the "everything else" (the capabilities we want to preserve) are *separable by token position*: the tags live at supervised positions, the content lives at the other positions. We therefore split the objective in two:

- **Tags**: the weighted cross-entropy to the supervised tag tokens, exactly as in the plain SFT recipe.
- **Everything else**: a per-token forward-KL to the *frozen base model* on content positions, so the student's distribution is anchored to the pretrained policy wherever there is no explicit tag supervision.

The teacher is the frozen original model itself conditioned on the same duplex sequence, so no second model and no rollouts are needed. Because the student is *base + LoRA*, the teacher forward is obtained with zero extra VRAM by temporarily disabling the adapters, and the teacher's top-$k$ next-token logits are precomputed once (offline cache), which removes the second forward pass entirely and makes the run no more expensive than SFT.

### B. Loss

With $\hat{p}_i$ the student distribution at position $i$, $p_i^{\text{base}}$ the frozen base distribution, $y_i$ the supervised tag and $w_i$ its weight,

$$\mathcal{L} = \underbrace{\frac{1}{N_{\text{sup}}} \sum_{i \in \text{tags}} w_i \, \text{CE}(\hat{p}_i, y_i)}_{\text{tag cross-entropy}} + \lambda \underbrace{\frac{1}{N_{\text{sup}}} \sum_{i \in \text{content}} D_{\text{KL}}\big(p_i^{\text{base}} \,\|\, \hat{p}_i\big)}_{\text{content anchoring}},$$

with $\lambda = 1.0$ chosen by a short sweep over $\lambda \in \{0.1, 0.3, 1.0\}$ scored on tag accuracy and KL drift.

### C. Practical details

Two implementation details mattered. First, *artifact masking*: in every `user finish speaking` item the first two *content* tokens of the response are left unsupervised, so the model is not forced to reproduce the teacher's opening tokens verbatim while the tag itself—predicted by the position before it—stays fully supervised. Second, *capability anchors*: 300 anchor prompts (150 function calling, 150 general QA) are mixed into the training set with `is_anchor` set, so the formats we intend to protect are also anchored by the KL rather than only by the base weights.

## VI. Approach C: an MTP-style protocol companion

### A. Separation of concerns

Instead of teaching the answer model the protocol, we train a *small companion* that predicts *only* the protocol stream on behalf of the bigger model, while the bigger model continues to do what it already does: produce answer text. This is the same division of labour as an MTP head [16]—a lightweight module alongside the main model—except that our companion predicts a different vocabulary: the duplex protocol, not the next answer token. Where no control token applies it emits nothing (an empty string that never enters the context). The companion is therefore the *controller* of the system: it decides turn starts and ends, barge-ins, backchannels and silences, without any voice-activity detection.

### B. Vocabulary and supervision

Let $\mathcal{T}$ be the companion's output vocabulary, a set of nine names: `no_tag`, `user_is_speaking`, `user_finish_speaking`, `user_interruption`, `user_backchannel`, `user_is_thinking`, `system_backchannel`, `system_take_floor` and `system_handover`, where `no_tag` is a label-only symbol that never enters the context. For every item $i$ of a dialogue the target is the tag carried by the *next* assistant item:

$$\text{target}(i) = \begin{cases} \text{special}(\text{item}_{i+1}) & \text{item}_i \text{ is user} \\ \texttt{no_tag} & \text{item}_i \text{ is assistant} \end{cases}$$

Each dialogue becomes one sequence with labels masked everywhere except at item boundaries. This gives the companion exactly the supervision the paper's model received, but as a *classification* of the protocol state rather than a generation task.

### C. Controller augmentations

The vanilla targets cover the paper's phenomena, but two behaviours are critical for a controller and under-represented, so the prep adds ∼946k records on top of the 10k main streams:

*a) Streaming-aware supervision.:* The runtime queries the companion while user words are still arriving, so every multi-token user chunk also yields partial-prefix records labelled `user_is_speaking`—except partials of an interrupting chunk, labelled `user_interruption`.

*b) Barge-in records.:* At random points inside every system turn the current assistant item is rendered *open* (speech in progress) followed by (a) the user's next real words → `user_interruption`, (b) `<|no voice|>` → `no_tag` (silence during speech means keep talking), or (c) a sampled pt-BR backchannel word → `user_backchannel`. This is what lets the companion distinguish "the user barged in" from "the user just said *aham*" from "the user paused".

*c) Silence semantics.:* The runtime inserts a `<|no voice|>` marker on ASR silence, so the data must define what silence means in every state: non-final chunk + silence → keep speaking; *final* chunk + silence → `finish` (turn complete—the case the original data never shows, since `finish` always directly follows the final chunk); partial final chunk + silence → keep speaking; and minimal single-utterance records [user chunk, `<|no voice|>`] at weight 5, without which the model never fires `finish` on a two-item context.

*d) Promptable-controller records.:* To make the controller *promptable*, a system instruction (a *rule*) is rendered as the first context item and the boundary after a trigger chunk is labelled `system_take_floor` (the system interrupts the user immediately) or `system_handover` (control passes to the big model). Synthetic scenarios use the real dictation shape—digit chunks with commas, the full alphabet as letters—so the companion learns the concept (number vs. letter, trigger vs. unrelated words) rather than memorized strings; negatives prevent spurious fires. About half of the

regular records additionally carry a generic instruction so the base controller behaviours survive instruction-conditioned training.

### D. Loss and inference

The loss is the same two-term objective as Section V: masked cross-entropy with per-token weights at item boundaries (retuned for the 9-way decision—finish = 3, interruption = 6, backchannel = 3, thinking = 2, speaking = 2, take_floor = 5, handover = 3, `no_tag` = 0.5), plus a per-record multiplier so rare controller patterns dominate the gradient, and a forward-KL to the frozen base on content positions with $\lambda = 0.3$. Preserving the companion's language ability is not cosmetic—the rule mechanism is an instruction-following behaviour.

At inference the companion is queried per event (new ASR words, silence timer, assistant sentence boundary); a tag fires only if its probability exceeds a per-tag threshold, and `user_interruption` is additionally suppressed when the last user item is the orchestrator-inserted `<|no voice|>` marker, so silence can never hard-stop the assistant. The context rendering must match training exactly: user items are always closed, even for partial content, and only an in-progress assistant item is rendered open.

## VII. Evaluation

### A. Protocol metric (teacher-forced)

Tag accuracy on a held-out duplex split: at each supervised tag position the argmax must equal the supervised special token. This is the metric the three approaches are usually compared on. We also report the mean per-token forward-KL between each student and the frozen base on the held-out content positions—a direct measure of "everything else stayed the same".

### B. Capability metrics

18 pt-BR function-calling prompts scored by native Qwen tool calls (greedy, tool-name plus required/expected arguments, digit-normalized) and 12 general instruction/QA prompts. Standard benchmarks: HellaSwag, ARC-Challenge and MMLU (2,000-example subsets), GSM8K (5-shot, 300), TruthfulQA (mc2) and IFEval (40) via lm-evaluation-harness 0.4.12, raw prompts without a chat template and identical configs across models. The companion is additionally evaluated on its own tag-stream held-out split with per-tag precision/recall/F1 and on hand-built runtime probes.

### C. Free generation: Full-Duplex-Bench

The metrics above are teacher-forced: at every supervised position the model is handed the correct prefix and only its next-token choice is scored. That is the right measure of whether the protocol was *learned*, but it says nothing about whether the model *uses* it when generating on its own, which is what a deployed duplex system needs. We therefore add a free-generation evaluation modelled on Full-Duplex-Bench (FDB) [3].

*a) Dimensions and metrics.:* FDB v1.0 scores four behaviours: *pause handling* and *backchannel* (the model must stay silent or briefly acknowledge: TOR ↓, backchannel frequency ↑, JSD against a ground-truth backchannel-time distribution ↓), *smooth turn-taking* (take the floor as soon as the user's turn ends: TOR ↑, latency ↓) and *user interruption* (yield, then answer the interrupting turn: TOR ↑, latency ↓, and an LLM-judged 0–5 relevance of the post-interruption reply). We use FDB's thresholds verbatim—a model utterance is a formal turn at $\geq$1 s or >3 words, a backchannel at $\leq$3 s and $\leq$3 words—and its interruption rubric verbatim. Each subset is generated at FDB v1.0's published size (216/137/119/55/200 scenarios, 727 per model), in pt-BR and in English.

*b) Text-domain adaptation.:* Our cascade decides turn-taking in text, so the harness replays the ASR stream as timed events rather than audio and converts text to seconds with a calibrated speaking rate (3.25 words/s, measured against the pocket-tts voice the live bridge uses and anchored to the rate implied by FDB's own released transcripts). Latency is the model's real inference compute between the user's turn ending and the model committing—the honest text-domain analogue of acoustic response latency, and the quantity that separates a tag-emitting LLM from a separate policy companion. Interruption relevance is judged by a local `Qwen3.5-9B` GGUF under FDB's own rubric; the judge is not GPT-4o, so that column ranks our models against each other rather than against the published absolute. All runs are greedy and every model is served through the same llama.cpp stack at the same quantisation.

*c) The input stream must match training.:* The models never see whole sentences. The micro-turn builder delivers each user turn as 1–7-**token** pieces (Section III), so we reuse that algorithm verbatim, including its mid-word cuts and leading-space artifacts, to generate the evaluated stream. This is not a detail, and the ablation in Section IX separates the two ways it matters. Our first harness deviated on two axes at once: it fed whole sentences as ASR pieces *and* joined several sentences into one turn. The single-factor arms show that the second axis is what distorted the turn-taking numbers—multi-sentence turns more than triple pause-handling TOR, because sentence-final ambiguity scales with pieces per turn—while the first axis breaks interruption recovery instead, because a one-piece interrupting turn is a shape the training builder never produces. We therefore generate every task's turns and pieces to the measured distribution of Section III.

*d) Every scenario is scored.:* FDB v1.0 consults its judge in exactly one dimension, so the paper-comparable column is the 200-sample interruption subset— that is the whole subset, not a subsample. Our harness additionally scores every scenario in which the model spoke ($\sim$400 per run) on a shared 0–5 relevance rubric, and scores the rest by rule (silence correct → 5, silence when a reply was due → 0), giving a whole-dataset quality column.

*e) The reference model.:* To anchor the comparison we download the upstream English DuplexCascade release, merge its PEFT state dict, quantize to 4-bit and serve it through

TABLE I
PROTOCOL AND CAPABILITY RETENTION (OUR PT-BR EVAL SETS).

| | base | **distill** | sft |
|---|---|---|---|
| Tag accuracy (%) | 0.0 | **96.4** | 98.0 |
| Function calling (%) | 100 | **100** | 55.6 |
| General instruction/QA (%) | 100 | **100** | 41.7 |
| KL drift vs base (nats) | 0.51 | 1.07 | 4.76 |

TABLE II
STANDARD BENCHMARKS (LM-EVALUATION-HARNESS 0.4.12; SUBSETS AS NOTED).

| Benchmark | base | **distill** | sft |
|---|---|---|---|
| HellaSwag acc-norm ($n$=2000) | 0.584 | 0.566 | 0.593 |
| ARC-Challenge acc-norm ($n$=2000) | 0.584 | 0.563 | 0.574 |
| MMLU ($n$=2000) | 0.705 | 0.698 | **0.660** |
| GSM8K exact ($n$=300, 5-shot) | 0.890 | 0.777 | 0.827 |
| TruthfulQA mc2 | 0.626 | **0.649** | 0.601 |
| IFEval loose / strict ($n$=40) | 0.730/0.651 | **0.762/0.683** | 0.603/0.524 |

the identical stack. Three conventions of its own codebase had to be matched, and each is documented because getting them wrong makes the reference look broken: it prepends BOS before the first micro-turn; it *speaks by emitting text* rather than a tag (its server streams any generated text to TTS and uses the tags only to stop); and it commits to the floor on the *silence* that follows speech rather than on the final word, so scenarios must include the trailing silence that real audio always has. Its tag vocabulary also differs ("talking" vs. "speaking", and no system-backchannel token), and is carried per-model.

## VIII. RESULTS

### A. Protocol vs. capability retention

Table I is the central result for the two in-model approaches. The distill model learns the protocol (96.4% tag accuracy, 1.6 pp behind SFT) while keeping function calling and general instruction-following at *base level* (100%), with a KL drift only slightly above the base floor. The plain-SFT baseline learns the protocol slightly better (98%) but regresses on both format-following capabilities and drifts $\sim 3.5\times$ further from the base distribution.

### B. Standard benchmarks

Table II reports the lm-evaluation-harness results. The distill model is within $\sim$1–2 pp of base on knowledge, commonsense and reasoning, *above* base on factuality and instruction following, and at base on the custom sets. The SFT baseline drops on MMLU, TruthfulQA, IFEval and the custom sets. The one regression of the distill model is GSM8K, discussed in Section XI.

### C. Companion: tag-stream accuracy and controller probes

Table III reports the companion's per-tag results on its held-out split (3,222 boundary decisions, restricted argmax over the tag vocabulary). The two rare but critical controller decisions—barge-in interruption and backchannel recognition—stay near-perfect, and the two promptable-controller tags are perfect. The small accuracy drop versus an earlier revision is the price of the instruction-conditioned training and the distill term.

TABLE III
PER-TAG RESULTS OF THE TRAINED COMPANION ON THE HELD-OUT DUPLEX SPLIT.

| **tag** | **P** | **R** | **F1** |
|---|---|---|---|
| no tag (emit nothing) | 1.00 | 0.95 | 0.973 |
| user is speaking | 0.99 | 0.92 | 0.953 |
| user finish speaking | 0.79 | 0.98 | 0.877 |
| user interruption | 0.95 | 0.99 | 0.969 |
| user backchannel | 1.00 | 0.98 | 0.99 |
| user is thinking | 0.90 | 1.00 | 0.949 |
| system take floor | 1.00 | 1.00 | 1.00 |
| system handover | 1.00 | 1.00 | 1.00 |
| accuracy | | 0.956 | |

Runtime probes on hand-built conversations exercise the query path with guards and thresholds applied: a mid-question user chunk $\rightarrow$ keep waiting; a finished question followed by silence $\rightarrow$ `finish` (fires the answer); a barge-in during assistant speech $\rightarrow$ `interruption`; "aham" during assistant speech $\rightarrow$ `backchannel` rather than an interruption; silence during assistant speech $\rightarrow$ keep talking. All are correct with probability $\geq 0.83$ at the decision point, the remainder being carried by the no-voice guard and the state machine. With a rule configured, a letter inside a digit dictation fires `system_take_floor` at 0.60–0.86, the same rule with digit chunks only does not fire, and without a rule the system tags never fire at all.

### D. Free-generation turn-taking

Table IV reports the FDB results for the three approaches, the untrained base as a control, and the upstream release measured with the same harness. Table V gives the same comparison in English, which is out-of-distribution for all three trained variants (§VIII-D0d).

*a) The harness reproduces the reference.:* The upstream release, measured here, lands on its own published numbers: pause handling 0.102/0.273 against the published 0.058/0.222, smooth turn-taking 0.975 against 0.832, interruption TOR 0.985 against 0.955, interruption relevance 3.92 against 4.02, and ATA 0.896 against 0.858. A re-implementation that reproduces the reference model's scores is measuring the same thing the reference measures, which is the strongest available check on everything else in these tables.

*b) All three approaches beat the reference on turn-taking accuracy.:* They reach ATA 0.973 (companion), 0.971 (SFT) and 0.955 (self-distillation)—every one above the upstream model's 0.896 and above the published 0.858—and all three sit at or better than the published figures on every TOR column. They get there differently, and that difference is the practical point of the paper: the companion has the best interruption relevance and the best candour-pause behaviour while a separate controller is queried before audio can be cut (stop latency 0.066 s), whereas SFT is an order of magnitude

TABLE IV

FULL-DUPLEX-BENCH, PT-BR (727 SCENARIOS PER MODEL). TOR AND LATENCY ARE BETTER WHEN LOWER; TOR, RELEVANCE AND QUALITY ARE BETTER WHEN HIGHER. "ORIGINAL" IS THE UPSTREAM ENGLISH RELEASE MEASURED WITH THE SAME HARNESS; THE PUBLISHED ROW IS ITS PAPER'S ENGLISH AUDIO RESULT AND IS A RANGE CHECK, NOT A BASELINE. QUALITY IS A WHOLE-DATASET 0–5 REPLY RUBRIC (SEE §VII-C).

| Model | Pause $\text{TOR}_{\text{syn}}$ | Pause $\text{TOR}_{\text{cand}}$ | BC TOR | Smooth TOR | Floor lat. (s) | Intr. TOR | Intr. relevance | Stop lat. (s) | **ATA** |
|---|---|---|---|---|---|---|---|---|---|
| Companion (MTP-style) | 0.073 | **0.000** | 1.000 | 0.975 | 0.33 | 0.990 | **4.36** | 0.066 | **0.973** |
| SFT (original recipe) | 0.080 | 0.032 | 1.000 | **1.000** | **0.25** | **0.995** | 3.56 | **0.035** | **0.971** |
| Self-distillation | 0.095 | 0.065 | 1.000 | 0.983 | 3.29 | 0.995 | 1.59 | 0.351 | 0.955 |
| Original DuplexCascade (upstream) | 0.102 | 0.273 | **0.436** | 0.975 | 1.02 | 0.985 | 3.92 | 0.323 | 0.896 |
| Base (no fine-tune, control) | 0.000 | 0.000 | 0.000 | 0.000 | – | 0.000 | – | 0.15 | 0.000 |
| *published DuplexCascade (EN audio)* | *0.058* | *0.222* | *0.218* | *0.832* | *1.72* | *0.955* | *4.02* | *1.23* | *0.858* |

TABLE V

FULL-DUPLEX-BENCH IN ENGLISH (SAME HARNESS, SAME SCENARIO SIZES). ENGLISH IS OUT-OF-DISTRIBUTION FOR ALL FOUR TRAINED MODELS—THEY ARE FINE-TUNED ON PT-BR ONLY—SO THIS MEASURES TRANSFER OF THE PROTOCOL, NOT LANGUAGE SKILL.

| Model | Pause $\text{TOR}_{\text{syn}}$ | Pause $\text{TOR}_{\text{cand}}$ | BC TOR | Smooth TOR | Floor lat. (s) | Intr. TOR | Intr. relevance | Stop lat. (s) | **ATA** |
|---|---|---|---|---|---|---|---|---|---|
| SFT (original recipe) | **0.051** | **0.000** | 1.000 | **1.000** | **0.27** | 0.995 | 4.08 | **0.031** | **0.986** |
| Companion (MTP-style) | 0.066 | **0.000** | 1.000 | **1.000** | 0.32 | **1.000** | **4.88** | 0.057 | **0.984** |
| Original DuplexCascade (upstream) | 0.095 | 0.102 | **0.345** | **1.000** | 1.16 | **1.000** | 4.35 | 0.251 | 0.951 |
| Self-distillation | 0.095 | 0.065 | 1.000 | 0.941 | 4.34 | 0.915 | 2.12 | 0.189 | 0.924 |
| Base (no fine-tune, control) | 0.000 | 0.000 | 0.000 | 0.000 | – | 0.000 | – | 0.15 | 0.000 |

faster to stop and to respond (0.035 s and 0.28 s) because one model does everything.

*c) What teacher forcing hid.:* The distill model is competitive on turn-taking accuracy but has by far the weakest interruption relevance (1.59 against 3.56 and 4.36) and is 13× slower to commit to the floor (3.29 s against 0.25 s). Teacher-forced tag accuracy (96.4%, Table I, 1.6 pp behind SFT) is blind to exactly this: both models put the right tag at the right position when given the correct prefix, but in free generation the distill model's continuation artifact—the known failure in which it opens by finishing the user's own sentence—degrades the reply and delays the turn. The live bridge masks it by stripping leading continuation punctuation; this evaluation does not, so the cost is visible. This is the concrete form of the exposure-bias gap between the protocol metric and deployed behaviour, and it is the strongest argument for evaluating turn-taking in free generation rather than at supervised positions.

*d) Cross-lingual transfer.:* All variants are fine-tuned on pt-BR only, and the protocol transfers: ATA is 0.986 (SFT), 0.984 (companion) and 0.924 (distill) on the English scenarios—SFT and the companion are as good or better in English than in pt-BR, with the companion's interruption relevance rising to 4.88. Only the distill variant degrades (0.955→0.924, interruption TOR 0.995→0.915), consistent with it being the most fragile of the three. The ranking of the approaches is stable across languages, which is a good sign that it reflects the models rather than the run.

The two language runs are parallel translations of the same synthetic scenarios, so the comparison is controlled—but it is not symmetric, and the asymmetry is worth naming because it produces an apparent inversion. The Portuguese stimuli are longer than their English originals, so the chunker yields more pieces: the same 727 scenarios produce 5.78 user pieces against 4.57, and every turn-taking rate is measured over those opportunities. The whole of the SFT inversion is therefore the pause term: pause TOR 0.051→0.080 and 0.000→0.032 give $-0.007-0.008$ of ATA, which is the entire 0.986→0.971 difference, while smooth-turn and interruption TOR are unchanged to three decimals. In absolute terms the pause term moves by a handful of scenarios out of 353. The model that changes most is the upstream English release, whose pause TOR doubles (0.099→0.207 over the two pause subsets): the direction a genuine cross-lingual effect would predict for a model trained in English, and the opposite of what the in-distribution reading of the trained variants would suggest. On real audio the same asymmetry appears far larger ($\S X$), for reasons that belong to the measurement rather than to the models.

*e) Backchannel is a data gap, not a capability gap.:* Every pt-BR variant scores BC TOR $= 1.000$ where the upstream model scores 0.436: they always take a formal turn during a monologue, while upstream backchannels. The cause is in the training data—across 10,249 sequences the tag `<|system backchannel|>` occurs **zero** times, and `<|user backchannel|>` only 1,262 times (0.09% of duplex tags)—so the token exists in the vocabulary but was never trained. That column is not evidence about these models and we do not interpret it. The same table also shows why a single headline number is not enough: the untrained control scores `chunk_accuracy` 0.85 on the turn-end decision—*higher* than every trained model—purely by never taking the floor. Our averaged turn-taking accuracy gates on the model demonstrably taking turns elsewhere and scores an unmeasured interruption as zero, which puts the control at 0.000.

### *E. Where the decisions go*

A diagnostic that turned out to explain the whole comparison: the two codebases commit to the floor at *different moments*. SFT, distill and the companion fire on the final

user chunk (rate 0.94–1.00) and essentially never on the post-turn silence; the upstream release fires on the silence (0.95–0.99) and almost never on the final chunk (0.013). Both conventions reach $\sim$1.0 smooth turn-taking and interruption TOR—they are inverses of one another, not better or worse—and this is why the upstream model measured as unable to take a turn at all until its trailing-silence cue was supplied. Meanwhile `pause_stayed_silent` is $\sim$1.000 for every trained model, original included: *nobody* speaks during a mid-turn pause. The differences between these models are about *when* they commit, not about whether they can hold.

## IX. Ablations

We isolate the three factors that the results above depend on. Each arm changes exactly one factor with respect to the main run—same 727 scenarios, seed, clock, decoding and serving stack—so the differences in Table VI are attributable to that factor alone.

*a) Turn length, not piece granularity, drives pause handling.:* The two axes of the input distribution had to be separated, and doing so corrected our own earlier account of the harness bug. Feeding whole sentences as ASR pieces (A1) leaves pause handling essentially unchanged—0.080 synthetic, 0.097 candour—while making the turns multi-sentence (A2) more than triples the synthetic-pause TOR to **0.307**. The cause is the number of sentence-final pieces per turn: the real data carries $\sim$1.5, A2 carries 3.1, and each additional sentence-final piece is another opportunity to fire `finish` early. What moves the pause metric is therefore *how much of the turn is still unheard*, not how the ASR slices it. Our first harness did both at once and we initially attributed the effect to the slicing; the single-factor arms show it belongs to the turn length.

*b) Piece granularity drives interruption recovery.:* The opposite holds for interruptions: A1 collapses interruption TOR from 0.995 to **0.000** while leaving every other column intact. A one-piece interrupting turn is off-distribution in a specific way—the training builder refuses to interrupt with a single-chunk turn, because that would produce two consecutive assistant messages and break strict alternation—so the model has never seen an interruption delivered as one long piece and does not re-take the floor afterwards. This is a concrete example of a training-data invariant that a benchmark can silently violate, and it is why the harness reproduces the builder's piece sizes rather than approximating them.

*c) Controller augmentations pay for themselves in precision.:* The companion ablates cleanly against its earlier revision (A3): without the promptable-controller records and the two extra tags, ATA falls from 0.973 to 0.954. The trade is instructive—A3 has *better* recall on the turn-end decision (1.000 vs. 0.958) but worse precision (0.680 vs. 0.698), and that precision loss is what shows up as worse pause handling (0.088/0.097 vs. 0.073/0.000). The augmentations make the controller more conservative about committing, which is exactly what the stay-silent tasks reward. Almost all of the missing behaviour is in the candour-pause column; the interruption and smooth-turn columns are unchanged or slightly better in the ablation, so the cost of the extra machinery is confined rather than global.

*d) The clock is not a sensitive parameter.:* Because durations in this harness come from a calibrated speaking rate rather than from audio, we recomputed every TOR column of the four trained models at 2.5, 3.0, 3.25 and 4.0 words/s—a $\pm 25\%$ band around the calibrated value. Every value is identical to three decimals at all four rates. The reason is that the decisions sit far from the duration gates: a six-word reply is 1.5–2.4 s at these rates, comfortably above the 1 s "formal turn" threshold, so the word-count branch of FDB's rule decides rather than the duration branch. The clock is therefore a documented approximation that does not drive any conclusion in this paper. The latency columns are measured inference compute and are unaffected by it by construction.

### A. The distilled model's prompt, and its degeneration

The arms above are text-domain. This one is on the real Portuguese audio (§X-A), and it changes one factor: the system prompt. The distilled model is trained on data that carries no system item at all, so the training distribution is an empty prompt; the prompt the shipped bridge uses is the one in `DuplexCascade/server.py`, which forbids the protocol tags ("*Responda apenas com a fala—não imprima nenhuma tag especial*"). We also measure the variant that forbids nothing, replacing that one clause with a permission to use the tags—the same edit that distinguishes the teacher's two prompts—so the model can express the protocol itself and the prompt is the only thing that changed.

*a) The prompt fixes the degeneration, and pays in commitment.:* With no prompt the distilled model answers in a degenerate register: its replies open with a stray non-Latin token in 18.8% of cases, they run to 34 words, and the interruption relevance is 1.56, against 3.10 for SFT and 3.19 for the companion on the same set. The production prompt removes almost all of it—the stray token drops to 4.8%, the replies to 14.7 words, the relevance more than doubles to 3.36, and the floor latency falls by an order of magnitude, 2.36 s to 0.22 s—and the backchannel and pause columns improve with it, because the model stops talking over monologues. What it costs is commitment: the model takes the floor 909 times instead of 1422 and its smooth-turn TOR falls from 0.378 to 0.143, so the guarded ATA collapses to 0.143 while the plain four-term mean is 0.550. The degeneration is thus a token-level artefact of the no-prompt setting rather than a property of the distillation, and it is fixable at the prompt level.

*b) Why the prompt and the priming belong together.:* The production prompt is what the bridge ships, but the bridge also *primes* the model with `<|user finish speaking|>` at the end of each user turn, and the prompt's ban on tags is why it has to: with the tags forbidden the model cannot say "the user has finished", so the turn decision has to come from somewhere else. The arm above isolates the prompt—decision mode, no priming—and is a diagnostic rather than a product configuration; the shipped product runs the prompt *and* the priming.

TABLE VI

ABLATIONS ON THE PT-BR FDB HARNESS. EACH ARM DIFFERS FROM ITS MAIN RUN IN ONE FACTOR. A1/A2 ABLATE THE SFT MODEL; A3 ABLATES THE COMPANION. TOR ↓ FOR THE PAUSE AND BACKCHANNEL COLUMNS, ↑ FOR SMOOTH AND INTERRUPTION.

| Arm | Factor changed | Pause $TOR_{syn}$ | Pause $TOR_{cand}$ | Smooth TOR | Intr. TOR | **ATA** |
|---|---|---|---|---|---|---|
| *SFT: input-stream and turn-length axes* | | | | | | |
| main | none (training pieces, training-sized turns) | 0.080 | 0.032 | 1.000 | 0.995 | **0.971** |
| A1 | user pieces = whole sentences | 0.080 | 0.097 | 1.000 | **0.000** | 0.706 |
| A2 | user turns = multi-sentence (2–4) | **0.307** | 0.056 | 1.000 | 0.995 | 0.908 |
| *Companion: controller augmentations* | | | | | | |
| main (v6) | full augmentation set, 9 tags | **0.073** | **0.000** | 0.975 | 0.990 | **0.973** |
| A3 (v5) | no promptable-controller set, 7 tags | 0.088 | 0.097 | **1.000** | **1.000** | 0.954 |

TABLE VII

THE DISTILLED MODEL WITH AND WITHOUT A SYSTEM PROMPT, ON THE REBUILT PORTUGUESE AUDIO (727 SCENARIOS EACH, READ BY FASTER-WHISPER LARGE-V3 IN PORTUGUESE). THE LAST THREE COLUMNS ARE THE DEGENERATION: THE SHARE OF REPLIES OPENING WITH A STRAY NON-LATIN TOKEN, THE MEAN REPLY LENGTH, AND HOW MANY TIMES THE MODEL TAKES THE FLOOR.

| Arm | Prompt | Pause cand TOR ↓ | BC TOR ↓ | Smooth TOR ↑ | Floor lat. (s) ↓ | Intr. relevance ↑ | ATA ↑ | ATA raw | Stray-token replies | Words/turn | Floor takes |
|---|---|---|---|---|---|---|---|---|---|---|---|
| main | none (training distribution) | 0.157 | 0.364 | 0.378 | 2.36 | 1.56 | 0.378 | 0.650 | 18.8% | 34.4 | 1422 |
| A4 | production prompt, tags permitted | **0.148** | **0.145** | 0.143 | **0.22** | **3.36** | 0.143 | 0.550 | **4.8%** | **14.7** | 909 |

TABLE VIII

ANCHORING STRENGTH VS. CAPABILITY RETENTION (PT-BR: 18 FUNCTION-CALLING AND 12 GENERAL INSTRUCTION/QA PROMPTS, GREEDY, IDENTICAL PROMPTS AND SCORING FOR EVERY ARM). $\lambda$=0 IS PLAIN SFT.

| Arm | Steps | $\lambda$ | FC acc. ↑ | General acc. ↑ |
|---|---|---|---|---|
| base (no fine-tune) | – | – | 1.000 | 1.000 |
| SFT (no KL term) | 3000 | – | **0.556** | **0.417** |
| distill | 300 | 0.1 | 1.000 | 1.000 |
| distill | 300 | 0.3 | 1.000 | 1.000 |
| distill | 300 | 1.0 | 1.000 | 1.000 |
| distill (shipped) | 2000 | 1.0 | 1.000 | 1.000 |

### B. Anchoring strength and capability retention

The retention result of Section VIII—distillation keeps function calling and instruction following where SFT loses them—rests on a single hyper-parameter, the anchoring weight $\lambda$. We vary it and re-measure the two capabilities that collapse under plain SFT, using the same prompts, decoding and scoring as the project's committed capability evaluation (Table VIII). All six arms were run again for this paper, and the three reference arms reproduce their committed values exactly—base 1.000/1.000, SFT 0.556/0.417, shipped distill 1.000/1.000—which is the reproducibility check on the one arm whose number the paper's claim depends on.

Retention behaves as a *step* in the presence of the anchoring term rather than a curve in its weight: removing it entirely costs more than half of both capabilities (1.000→0.556 and 1.000→0.417), while a ten-fold change in $\lambda$ from 0.1 to 1.0 changes nothing measurable. The practical consequence is that $\lambda$ does not need to be tuned to protect capabilities—the reported $\lambda$=1.0 is safe on this axis, and the real cost of a large $\lambda$ lies elsewhere, in the free-generation behaviour of Section XI and the GSM8K trade-off.

One caveat is inherent to the runs we have: the $\lambda$ arms are 300-step (the sweep the project trained) while plain SFT is 3000-step and the shipped distill is 2000-step at $\lambda$=1.0. The ablation therefore establishes that $\lambda$ is flat over 0.1–1.0 at a matched 300-step budget, and that retention still holds at full budget for $\lambda$=1.0; it does not measure a $\lambda$ sweep at full budget, which is the experiment that would rule out a small-$\lambda$ degradation appearing only with longer training.

## X. REAL-AUDIO EVALUATION

Everything up to here measures turn-taking in text: the stimuli are generated dialogues and a word rate turns a reply into seconds. Both approximations can be removed, so we remove them, and replay the benchmark's own 727 WAVs (3.6 h of real speech) through the deployed speech stack, scoring the audio that comes out with the benchmark's own evaluators (`fdb_bench/audio/`; design, code and caveats in its README).

*a) What the audio path replaces.:* The user's speech is the benchmark's audio; what the model reads is `faster-whisper medium` output delivered at the deployed service's own cadence (a 0.6 s partial interval, one beam-5 pass when 0.6 s of silence ends the utterance, `Eos` at the end of the stream); the turn-end invitation comes from silero VAD—the detector the benchmark's own scripts run on model output—instead of from an annotation; the reply is pocket-tts audio, so every duration gate is measured rather than estimated; barge-in truncates real playback; and scoring is upstream's `eval_*.py` reading `output.json` from `parakeet-tdt-0.6b-v2`, upstream's own recipe. The only substitution left is the interruption judge, as before.

TABLE IX
THE UPSTREAM RELEASE ON REAL FDB V1.0 AUDIO, SCORED BY UPSTREAM'S OWN EVALUATORS, AGAINST ITS PUBLISHED TABLE 1 ROW. ARROWS GIVE THE GOOD DIRECTION.

| Metric | Published | This run |
|---|---|---|
| Pause $TOR_{syn}$ ↓ | 0.058 | 0.088 |
| Pause $TOR_{cand}$ ↓ | 0.222 | 0.190 |
| Backchannel TOR ↓ | 0.218 | 0.182 |
| Backchannel frequency ↑ | 0.009 | 0.007 |
| Backchannel JSD ↓ | 0.949 | 0.964 |
| Smooth turn-taking TOR ↑ | 0.832 | 0.630 |
| Smooth floor latency (s) ↓ | 1.724 | 1.483 |
| Interruption TOR ↑ | 0.955 | 0.945 |
| Interruption relevance ↑ | 4.016 | 3.667 |
| Interruption latency (s) ↓ | 1.225 | 1.457 |
| **ATA ↑** | **0.858** | **0.824** |

*b) Three things the real data forced:* , each by measurement rather than assumption. The protocol's `<|no voice|>` ticks are load-bearing: with text pieces only, the upstream checkpoint never takes the floor at all, answering "wait" at every decision including after turn ends. The deployed 2.0 s commit window is a user-interface setting—it stops the browser's partial-text bubble revising a word, and the bridge never feeds partials to the LLM—and it starves a duplex loop: partial lag 2.41 s instead of 1.04 s, and utterances shorter than the window deliver nothing before the turn ends. And the deployed silence gate ($\texttt{rms} \geq 0.002$) is calibrated for a microphone whose idle state is digital silence, so on real conversation—RMS 0.006–0.24—it calls 94% of a file speech and never fires a turn end; the gate is therefore silero.

*c) The reference model reproduces its published row.:* Table IX puts the upstream release's published numbers beside the same model measured here on real audio with upstream's evaluators. Ten of eleven columns land in range, and the reference's ATA is within 0.034 of its published value. That is a much stronger statement than the text-domain check in §VII-C could make, because the audio path also exercises upstream's own ASR, its own duration gates and its own silence handling rather than a port of their rules.

Two honest qualifications. The relevance column is judged by the local 9B model answering upstream's rubric, not by GPT-4o, so it ranks models rather than reproducing an absolute. And smooth turn-taking TOR is the column that does *not* reproduce (0.630 against 0.832): it is not a missing invitation—all 119 scenarios received a turn end—but a failure mode, in all 43 failures the model answering "user is speaking" at every one of up to twelve silence ticks. It locks into a wait pattern on fragmentary real conversation, which the synthetic dialogues of §VII-C never presented.

*d) The text-domain ranking does not survive real audio.:* In text all three pt-BR variants beat the upstream model (ATA 0.986/0.984/0.924 against 0.951); on audio all three lose to it, and the text-domain winner—the companion—is the worst of the four (Table X). The cause is the invitation, and it is visible only once the two harnesses are compared per invitation rather than per scenario (Table XI). In text an invitation is the annotated turn end of a clean transcript; in audio it is silero firing on a real conversation gap, 3.7 per scenario and mostly mid-turn, where the user's last words are fragments. The upstream English model reads those as incomplete and waits—it takes the floor at 7.9% of audio invitations and 11.9% of text ones—while the pt-BR models jump from about 8.6% in text to 29–44% in audio. In a language they were never tuned for, they cannot tell an unfinished turn from a finished one, so they claim the floor at every second or third gap. Their scenario-level pause numbers are therefore not the same quantity as the text-domain ones, and the audio numbers are the trustworthy ones.

*e) The companion inverts.:* It is the most eager at English conversation gaps (pause TOR 0.920/0.829) and simultaneously the best at what it says once it takes the floor: interruption relevance 4.40 against the published 4.016, floor latency 0.61 s against the published 1.72 s, and the only model whose backchannel behaviour the metric rewards (frequency 0.111, JSD 0.722, against 0.009/0.949 published). That is the architecture's claim showing up in the audio—a dedicated timing head, an untouched content model—with the timing head transferring badly to English and the content model not having to.

*f) Language of the reply.:* The audio run adds a constraint the text run did not have: the benchmark's own scorer is English-only, so a Portuguese reply reaches the relevance judge as a pseudo-English transcript. We measure it with the local model rather than with a heuristic: the Qwen3.5-9B that answers the benchmark's interruption rubric labels every assistant turn `pt`, `en`, or another language (Table XII). On the English set, SFT answers in Portuguese for 26% of its turns and the other three models answer entirely in English; SFT's interruption relevance of 3.09 against the companion's 4.40 is partly that—a quarter of its replies are in the wrong language for the judge. The split is systematic rather than accidental: SFT is trained only on pt-BR duplex data, whereas distillation is anchored to the base model everywhere except the tags, and the companion's content comes from the stock model.

### A. The same benchmark in Portuguese

The three variants are fine-tuned on pt-BR duplex data, so a Portuguese re-voicing of the benchmark is their *in-distribution* set and the English recording is out-of-distribution for them—and the comparison between the two is the one that says whether the protocol generalises at all. Building that set honestly took a repair, because the first version inverted the comparison and we only found out by auditing the evaluation rather than the models.

*a) The re-voicing has to keep the recording's shape.:* The Portuguese set is the benchmark's own audio, segmented with the benchmark's own detector, translated, spoken by the synthesiser and put back on the timeline. The obvious way to put it back—time-fit each translated span into the English span it replaces and pad the remainder with silence—is wrong, and the reason is mechanical: the pad is silence *inside* the user's turn, the harness's 0.6 s silence gate fires on it, and the model

TABLE X

THE FOUR MODELS ON REAL FDB V1.0 AUDIO (727 SCENARIOS EACH), SCORED BY UPSTREAM'S EVALUATORS. "PAPER CONV." IS THE PAPER'S ATA RECONSTRUCTED AS THE MEAN OF FIVE ORIENTED TOR COLUMNS, BACKCHANNEL INCLUDED—IT REPRODUCES BOTH PUBLISHED ROWS EXACTLY; OURS IS THE PROJECT'S FOUR-TERM CONVENTION WITH ITS TWO GUARDS (§VII-C).

| Model | Pause $TOR_{syn}$ ↓ | Pause $TOR_{cand}$ ↓ | BC TOR ↓ | Smooth TOR ↑ | Floor lat. (s) ↓ | Intr. TOR ↑ | Intr. relevance ↑ | Intr. lat. (s) ↓ | **ATA** ↑ |
|---|---|---|---|---|---|---|---|---|---|
| *published DuplexCascade* | *0.058* | *0.222* | *0.218* | *0.832* | *1.72* | *0.955* | *4.02* | *1.23* | *0.858* |
| Original DuplexCascade (upstream) | 0.088 | 0.190 | 0.164 | 0.630 | 1.48 | 0.945 | 3.66 | 1.46 | **0.824** |
| Self-distillation | 0.219 | 0.264 | 0.345 | 0.697 | 3.80 | 0.605 | 1.47 | 2.71 | 0.704 |
| SFT (original recipe) | 0.825 | 0.713 | 0.673 | **0.866** | **0.34** | 0.930 | 3.09 | **0.65** | 0.564 |
| Companion (MTP-style) | 0.920 | 0.829 | 0.782 | 0.891 | 0.61 | 0.935 | **4.40** | 1.05 | 0.519 |

TABLE XI

INVITATIONS ON REAL AUDIO. "P(FLOOR)" IS THE PROBABILITY OF TAKING THE FLOOR GIVEN AN INVITATION—THE PER-SCENARIO TOR OF TABLES V AND X IS THAT RATE TIMES THE NUMBER OF INVITATIONS, WHICH IS WHY THE TWO HARNESSES CANNOT BE COMPARED ON THE SCENARIO FIGURE ALONE.

| Model | P(floor) text | P(floor) audio | P(floor) audio, pt set |
|---|---|---|---|
| Original DuplexCascade (upstream) | 0.119 | **0.079** | 0.129 |
| Self-distillation | 0.081 | 0.294 | 0.460 |
| SFT (original recipe) | 0.086 | 0.397 | 0.527 |
| Companion (MTP-style) | 0.086 | 0.441 | 0.521 |

is invited to take the floor in the middle of a sentence. In the first build 30% of the segments carried such a pad, and the turn-end invitations that fell in silence the benchmark does not annotate as a pause went from 58 on the English recording to 303 on the Portuguese one, over the same 353 scenarios. Every model answered more often because of it, the English-trained upstream release included, so the artefact could not be a language effect.

The set is therefore rebuilt with nothing stretched and nothing padded: the Portuguese speaker keeps their own speed, the gap between two consecutive spans keeps its length, and the annotations are re-timed with the same map. A structural gate then compares the result with the recording it was made from—speech fraction, spans, gaps of 0.6 s or more, and silence outside the benchmark's own pause windows—and the rebuilt set matches it (speech fraction 0.549 against 0.583, long gaps 0.94 against 0.98 per scenario, phantom silence 8.51 s against 8.44 s). The gate is what caught a second-order version of the same defect: the synthesiser pauses for longer than 100 ms at punctuation, which split the user's turns in two, so the spans now have to occupy their slots the way the recorded spans did.

*b) One recogniser, two languages.:* The benchmark's scorer is English-only and the two synthesised voices differ by 12.6 dB, so every rendered reply is peak-normalised before it is read—in both sets—and the Portuguese set is read with `faster-whisper large-v3` in Portuguese while the English set keeps the benchmark's parakeet for the published comparison. The language comparison below reads *both* sets with the same recogniser and changes only the language setting, because two recognisers measure two things.

*c) Out-of-distribution no longer beats in-distribution.:* On the rebuilt set the companion is level ($+0.005$), the distilled model's real gap is $-0.032$ (its $-0.304$ is the guard's cliff, not a Portuguese-capability gap), and the upstream release points the other way, as an English-trained model should. What remains is one asymmetry of the re-voicing itself: on the synthesised Portuguese speech every model—the upstream release included—takes the floor more often than on the English recording, which is what moves SFT's backchannel column by $+0.255$ and the upstream release's pause columns by $+0.12$. That is a property of the re-voiced audio, not of the models' Portuguese.

*d) The distilled model's degeneration is a token, not a language.:* On the English set the language judge calls 28% of the distilled model's replies Korean. They are not: the reply opens with a stray token (*!*) and the body is English—61% of its English replies contain such a token and 61% open with one, against **zero** for every other model. Strip the leading token and the model is answering in its own language 97% of the time (Table XII). On the Portuguese set the same artefact is milder (19% of replies) and the production prompt removes most of it (§IX-A), which is what makes it a fixable problem rather than a property of the distillation.

*e) Caveats.:* These rows are a pipeline measurement, not a live session: response latency excludes network jitter and audio-device buffering, one seed per configuration, and the Portuguese set is a re-voicing rather than a native recording: the synthesised speaker is not the one who produced the pauses, and its transcript is a machine translation of a machine transcript (§X-A). Replies that begin after the input ends are audible here only because the render carries a 4 s tail—upstream's own outputs are exactly input-length—and the trace keeps both readings (`spoke` versus `audible_within_input`); 2.8% of the upstream's replies and 5.9–16.8% of the pt-BR models' fall into that class, which is the deployed cadence's cost rather than the models'.

## XI. DISCUSSION AND LIMITATIONS

*Why the content-KL works.* The tags live at supervised positions and are predicted with the original high weights, so they are learned essentially as before. Every other position is anchored to the frozen base, which is exactly the behaviour we want to preserve. The mechanism is a single-stage, per-token analogue of the "re-invoke" two-stage procedure [11] and of SDFT's on-policy anchoring [2], but it requires neither rollouts nor a reward.

*Free-generation cost of the anchoring term.* The content-KL preserves capabilities but, at $\lambda$=1.0, it also anchors the model

TABLE XII

REPLY LANGUAGE ON REAL AUDIO, JUDGED BY THE LOCAL MODEL. EVERY ASSISTANT TURN IS LABELLED PT, EN OR ANOTHER LANGUAGE BY THE QWEN3.5-9B THAT ANSWERS THE BENCHMARK'S INTERRUPTION RUBRIC. THE "MAIN LANGUAGE" COLUMN IS THE SAME JUDGEMENT ON THE SAME REPLY WITH A STRAY LEADING TOKEN REMOVED, AND THE LAST COLUMN COUNTS NON-LATIN TOKENS DIRECTLY, WITH NO JUDGEMENT CALL—THE DISTILLED MODEL'S REPLIES OPEN WITH ONE (§IX-A).

| Set | Model | pt | en | other | main language | non-Latin (any) | (opening) |
|---|---|---|---|---|---|---|---|
| EN | Original DuplexCascade (upstream) | 0.0% | 100.0% | 0.0% | — | 1.0% | 0.0% |
| EN | SFT (original recipe) | 25.6% | 74.4% | 0.0% | — | 0.1% | 0.0% |
| EN | Self-distillation | 0.0% | 72.0% | **28.0%** | 97.1% en | **61.1%** | **60.6%** |
| EN | Companion (MTP-style) | 0.0% | 100.0% | 0.0% | — | 0.0% | 0.0% |
| PT | Original DuplexCascade (upstream) | 83.5% | 15.6% | 0.9% | — | 0.7% | 0.1% |
| PT | SFT (original recipe) | **98.6%** | 1.4% | 0.0% | — | 0.0% | 0.0% |
| PT | Self-distillation | 79.0% | 9.7% | 11.3% | 88.9% pt | 19.0% | 18.8% |
| PT | Companion (MTP-style) | 94.6% | 5.4% | 0.0% | — | 0.0% | 0.0% |
| PT | Self-distillation, production prompt | 81.4% | 12.6% | 5.9% | 82.9% pt | 7.6% | 4.8% |

TABLE XIII

THE FOUR MODELS ON THE PORTUGUESE RE-VOICING OF FDB V1.0 (727 SCENARIOS EACH), READ BY FASTER-WHISPER LARGE-V3 IN PORTUGUESE. "ATA RAW" IS THE PLAIN FOUR-TERM MEAN; THE GUARDED COLUMN IS THE PROJECT'S CONVENTION, WHOSE 0.5 CLIFF (SMOOTH TOR BELOW 0.5 COLLAPSES ATA TO SMOOTH) TURNS *late* INTO *mute*. THE LAST ROW IS THE DISTILLED MODEL IN ITS PRODUCTION CONFIGURATION (§IX-A).

| Model<br>ATA raw | Pause $TOR_{syn}$ ↓ | Pause $TOR_{cand}$ ↓ | BC TOR ↓ | Smooth TOR ↑ | Floor lat. (s) ↓ | Intr. TOR ↑ | Intr. relevance ↑ | Intr. lat. (s) ↓ | **ATA** ↑ |
|---|---|---|---|---|---|---|---|---|---|
| Original DuplexCascade (upstream)<br>0.740 | 0.234 | 0.315 | **0.182** | **0.639** | 1.31 | **0.870** | 1.94 | 0.97 | **0.740** |
| SFT (original recipe)<br>0.494 | 0.883 | 0.843 | 0.909 | 0.849 | **0.26** | 0.855 | 3.10 | **0.51** | 0.494 |
| Self-distillation<br>**0.650** | **0.212** | 0.157 | 0.364 | 0.378 | 2.36 | 0.590 | 1.56 | 3.85 | 0.378 |
| Companion (MTP-style)<br>0.520 | 0.730 | 0.806 | 0.836 | 0.782 | 0.97 | 0.835 | 3.19 | 1.10 | 0.520 |
| Self-distillation, production prompt<br>0.550 | 0.380 | **0.148** | **0.145** | 0.143 | 0.22 | 0.585 | **3.36** | 0.48 | 0.143 |

TABLE XIV

IN-DISTRIBUTION AGAINST OUT-OF-DISTRIBUTION, READ BY THE SAME RECOGNISER WITH ONLY THE LANGUAGE SETTING CHANGING. PORTUGUESE IS THE IN-DISTRIBUTION SET FOR THE THREE TRAINED VARIANTS, SO A NEGATIVE Δ MEANS THE MODEL DID *less* WELL ON ITS OWN LANGUAGE; THE UPSTREAM RELEASE IS ENGLISH-TRAINED, SO FOR IT THE SIGN IS EXPECTED TO POINT THE OTHER WAY. THE LAST FOUR COLUMNS SAY WHERE THE DIFFERENCE LIVES.

| Model | ATA EN (OOD) | ATA PT (ID) | Δ ATA | Δ raw | Δ pause cand | Δ backchannel | Δ smooth |
|---|---|---|---|---|---|---|---|
| Original DuplexCascade (upstream) | 0.821 | 0.740 | −0.081 | −0.081 | +0.120 | +0.000 | +0.008 |
| SFT (original recipe) | 0.556 | 0.494 | −0.062 | −0.062 | +0.116 | +0.255 | −0.008 |
| Self-distillation | 0.682 | 0.378 | −0.304 | −0.032 | −0.139 | −0.036 | −0.311 |
| Companion (MTP-style) | 0.515 | 0.520 | +0.005 | +0.005 | −0.032 | +0.073 | −0.109 |

to the teacher's content *generation style*, and the teacher was occasionally autocompleting the user turn when its auxiliary trigger asked for a short answer. The distillation therefore inherits that continuation artifact, which shows up in Table IV as the worst interruption relevance and the slowest floor commitment of any model measured, even though its teacher-forced tag accuracy is 96.4%. The fix we planned is at the data level, not the loss: regenerate the teacher content with a corrected trigger ("responda de forma natural e completa … não se repita") and retrain, so the artifact is absent from the anchored distribution rather than stripped at inference. The FDB harness now provides the measurement that tells us whether that retrain succeeded—relevance and stop latency, not tag accuracy.

*The GSM8K caveat.* The distill model drops 11 pp below base on GSM8K, a long-form multi-step reasoning task. This is the honest tradeoff of a strong anchoring term ($\lambda$=1.0): content distributions are heavily pinned to the base, and on 300 samples the 4B base's fragile chain-of-thought is sensitive to any distributional shift. Two natural follow-ups are a lower $\lambda$ (our sweep did not include a math benchmark) and the EMA-teacher or on-policy variants, which track the student's progress and may preserve adaptability on reasoning while still reducing forgetting.

*Turn size is an experimental variable.* Two generations of source dialogues exist in this project—short-only with a fixed 10-token assistant micro-turn, and the short/long/mixed mixture with 10–48-token chunking used here—and the assistant-side length distribution has a heavy right tail (median 5 words, p90 12, max 37). Because FDB's turn-taking gates are durations, that distribution moves the metric: a model trained on longer bursts crosses the formal-turn threshold

with fewer words. Benchmarking a duration-gated protocol therefore requires reporting the turn-length distribution on both sides, and we do so.

*What the companion buys and what it costs.* The companion makes turn-taking a reusable, attachable module: a 0.8B sidecar trained in hours on one GPU, swappable across big models with no retraining, and it supports instruction-conditioned interrupts that no single-model fine-tune provides. The costs are real: one extra model in the loop (stop latency 0.066 s versus 0.035 s for SFT), a learned turn-end signal that depends on the ASR punctuating its output—unpunctuated transcripts delay the `finish` decision—and a controller whose instruction-conditioned behaviours must be explicitly mixed with generic instructions so the base behaviours survive.

*Limitations.* Our free-generation harness is text-domain: durations come from a calibrated speaking rate rather than from audio, latency is inference compute rather than acoustic end-to-end latency, and the judge is a local 9B model, so the relevance column anchors our models against each other but not against the published GPT-4o absolute. The published row is English audio and our rows are pt-BR text; we use it as a range check rather than a baseline. All results are from a single seed per configuration, and the source dialogues are synthetic. Finally, one FDB dimension is unanswerable on this data by construction, as discussed above.

## XII. Conclusion

We compared three ways to give a conversational LLM the duplex-cascade micro-turn protocol, on one pt-BR pipeline and with one harness.

Plain **SFT** learns the protocol best (98% tag accuracy) but regresses on function calling and general instruction-following; it remains the fastest model at runtime and, in free generation, the best all-round turn-taker of the three. **Self-distillation** splits the objective by token position—weighted CE on the tags, forward-KL to the frozen base everywhere else—and retains base-level capabilities and knowledge without RL or rollouts, at no extra cost over SFT with an offline teacher cache. It is competitive on turn-taking accuracy but inherits the teacher's generation style, which costs reply relevance and latency in free generation—a cost teacher-forced tag accuracy (96.4%) cannot see. The **MTP-style companion** separates the two skills instead of balancing them: a 0.8B sidecar predicts only the control stream and drives an untouched big model, reaching 95.6% tag-stream accuracy with 0.97/0.99 F1 on barge-in and backchannel decisions and supporting instruction-conditioned interrupts, at the price of one extra model in the loop.

Evaluated in free generation with Full-Duplex-Bench, all three beat the published turn-taking accuracy, and re-measuring the upstream DuplexCascade release with the same harness reproduces its published numbers—which is what makes the comparison trustworthy. Three lessons generalize beyond this system: a protocol metric should be paired with a free-generation metric whenever the protocol is *used* rather than merely reproduced; a generative benchmark inherits the training input distribution, so that distribution is an experimental variable that must be reported (our single-factor ablations move pause-handling TOR by $4\times$ and interruption TOR to zero on the same scenarios); and a metric can be unanswerable by construction, as with the backchannel dimension here, where the required tag never appears in the training data. Per-token KL-anchoring to a frozen base remains a practical, RL-free anti-forgetting mechanism; the work here makes its cost measurable and localizes it to the teacher's generation style rather than its knowledge.

## Use of Generative AI

This manuscript was prepared with the assistance of generative artificial intelligence. All experimental results and the associated code were reviewed and verified by the author.

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